



Italian
Medical
Touch

MANUAL DE SERVICIO

WRITTEN
BY

MANUEL SACCANI
(Technical Service Manager)



Tabla de contenido

Instrucciones generales de seguridad.....	5
MEDIDAS SANITARIAS DE INTERVENCION.....	6
SOLUCION DE PROBLEMAS.....	7
PIEZAS O PARTES DE REPUESTO	8
PIEZA DE REPUESTO.....	8
DESCRIPCION	8
REF #	8
PIEZA DE REPUESTO.....	9
DESCRIPCION DE PIEZA DE REPUESTO.....	9
REF #	9
REEMPLAZO DE UNIDAD DE BOMBEO.....	10
PASOS PARA LA REPARACION.....	10
APERTURA DEL EQUIPO.....	10
DESCONECCION DE CONEXIONES ELECTRICAS Y NEUMATICAS.....	10
EXTRACCION DE ELECTROBOMBA Y EXTRACCIO DE UNIDAD DE BOMBEO.	10
RESTAURANDO EL DISPOSITIVO	11
NOTAS	11
Fig.1 – COMPONENTES DE LA UNIDAD DE BOMBEO.....	11
REEMPLAZO DE UNIDAD DE MOTOR	12
PASOS PARA LA REPARACION.....	12
APERTURA DEL EQUIPO.....	12
Desprendimiento de conexiones eléctricas y neumática.....	12
REEMPLAZO DE LA ELECTROBOMBA Y MOTOR.....	12
RESTAURACION DEL DISPOSITIVO.....	13
NOTES	13
Fig.2 – ELEMENTOS DE MOTOR.....	13
REEMPLAZO TODA ELECTROBOMBA.....	14
PASOS REPARACION.....	14
DESMONTAJE DE CONEXIONES ELECTRICAS Y NEUMATICAS.....	14
EXTRACCION Y REPARACION ELECTROBOMBA COMPLETA.....	14
RESTAURACION DEL DISPOSITIVO.....	14
NOTES	14
Fig.3 – ELECTROBOMBA.....	15
REEMPLAZO DE SOPORTE ANTIVIBRANTE DE ELECTROBOMBA.....	16
PASOS.....	16
DESCONEXION DE CONEXIONES ELECTICAS Y MECAN.....	16
EXTRACCION Y REEMPLAZO DE SOPORTE ANTIVIBRANTE.....	16
RESTAURACION DEL DISPOSITIVO.....	16
REEMPLAZANDO LA UNIDAD REGULADORA.....	17
PASOS.....	17
APERTURA.....	17
DESCONECTAR LA CONEXION NEUMATCA Y SUSTITUIR LA UNIDAD REGULADOR.....	17
RESTAURANDO EL DISPOSITIVO.....	17
Fig.4 – ELEMENTOS DE UNIDAD REGULADORA.....	17
REEMPLAZANDO EL CAPACITOR.....	18
PASOS PARA LA REPARACION.....	18
APERTURA.....	18
RESTAURANDO EL DISPOSITIVO.....	18

REEMPLAZANDO EL INTERRUPTOR.....	19
PASOS PARA LA REPARACION.....	19
APERTURA.....	19
Desenchufando las conexiones eléctricas y reemplazando el INTERRUPTOR	19
RESTAURACION DEL DISPOSIT.....	19
RESTAURACION DE LA BASE DEL FUSIBLE.....	20
PASOS PARA LA REPARACION.....	20
APERTURA.....	20
DESENCHUFE DE CONEXIONES ELCTRICAS Y REEMPLAZO DE SOCATE DEL FUSIBLE.....	20
RESTAURANDO EL DISPOSITIVO.....	20
REEMPLAZO DE LA UNIDAD MEDIDORA DE VA.....	21
PSOS PARA LA REPARACION.....	21
APERTURA.....	21
NOTAS.....	21
Fig.5 – ELEMENTOS DE LA UNIDAD DE MEDICION DE VACIO.....	22
REEMPLAZO DE LA MANIJA.....	23
REPAIRING STEPS	23
APERTURA.....	23
REEMPLAZANDO LA MANIJA.....	23
RESTAURANDO EL DISPOSITIVO.....	23
REEMPLAZANDO LA PARTE INFERIOR.....	24
PASOS.....	24
APERTURA.....	24
DESCONECTANDO LAS CONEXIONES ELECTRICASA Y NEUMATICAS.....	
Desenchufando la electrobomba y reemplazando la parte inferior del dispositivo	24
RESTAURANDO EL EQUIP.....	24
FIG. 6 – NEW ASKIR 30 REF RE 310100/02 ELECTRIC DIAGRAM	25
FIG. 7 – NEW ASKIR 30 REF RE 310100/02 (RIF. ED.AS._04.0010-2) EXPLODED VIEW.....	26

INSTRUCCIONES GENERALES DE SEGURIDAD

**Lea atentamente todas las instrucciones de este manual cuando trabaje con este producto;
Cualquier intervención debe cumplir estrictamente con las instrucciones del fabricante.**

Cualquier intervención de reparación y / o mantenimiento debe ser realizada por personal calificado.

**CA-MI consideraba solo personal calificado con el certificado específico de calificación,
publicado solo después de un curso especial de CA-MI.**

**CA-MI no se responsabiliza de ningún daño o problema, por parte de personas o personas,
derivado de intervenciones que no cumplan estrictamente con este manual de instrucciones de
operación y mantenimiento.**



**Antes de cualquier intervención en el dispositivo, asegúrese de que el equipo esté
desenchufado de la fuente de alimentación.
Para mayor seguridad, asegúrese de desenchufar el cable de alimentación de la toma
eléctrica.**



**Lea atentamente el párrafo MEDIDAS SANITARIAS DE INTERVENCIÓN y siga estrictamente
todas las instrucciones que aparecen en él.**

MEDIDAS SANITARIAS DE INTERVENCIÓN

- a) Any kind of maintenance and/or repairing intervention must be carried out in a close dedicated space and not in contact with the external environment.**
- b) It is necessary to wear adequate protective clothes (disposable latex gloves, surplice, disposable protective mask) before any intervention on the device.**
- c) If the device results or is suspected to be contaminated refer to points d),e),f),h).**
- d) The accessories or part of them which are contaminated must be replaced and stored as described at point e). The repairing can cause contamination both to the technician and to the final user.**
- e) All the replaced parts (vacuum pump, motor unit, bowls, accessories, pipes etc...) suspected to be bacterially contaminated must be treated and stored as hospital potentially contaminating stuff.**
- f) All CA-MI equipments presents two safety devices preventing those kind of failures (a “too full” device and an antibacterial and hydrophobic filter). Any penetration of contaminants in the product or in the pump unit arises from the improper use of the product itself (non-use of the antibacterial filter, non-substitution of the filter itself, etc.). In this case there could be a very high risk of bacterial contamination for all the users after the technical assistance intervention. The repairing is therefore highly UNADVISABLE and CA-MI must not be held liable for any eventual consequences.
In this case CA-MI will replace the whole equipment with another new one (see point h).
If the user wants to go on anyway, he has to consider what indicated at point c) and act consequently. Replacing all the tubes and the internal pipes is more advisable than repairing the contaminated vacuum pump. It must be replaced in the whole and thrown away as described at point e).**
- g) When there is no internal contamination, technical assistance interventions on the products must follow the instructions at points a) and b).
Before operating remove the accessories (disposable bowl or pocket, suction pipe and used pipes) as described at point e). After the intervention disinfect the external framework with a specific chemical preparation.**
- h) In case of internal contamination, due to an improper use of the product, or in case of failure of operating encountered after the end of the warranty period, you must communicate to CA-MI all the information about product’s BATCH NUMBER and SERIES NUMBER and discard the product itself as described at point e). CA-MI will replace it with another new one at the cost of euros.....**

For the remaining cases (break up, failure of operating due to a manufacturing defect etc...) CA-MI will replace the equipment with another new one.

SOLUCION DE PROBLEMAS

SOLUCIÓN DE PROBLEMAS

Defecto típico	Identificando el fallo	SOLUCIÓN DE PROBLEMAS
LA UNIDAD NO ENCIENDE: - El interruptor luminoso no se enciende y el motor no arranca - El interruptor se enciende pero el motor no arranca. - No gira y vibra ligeramente.	- Cable de alimentación interrumpido - Los fusibles interrumpieron dos para cortocircuitar o cambiar los cambios de voltaje - Cable interno desenchufado de la toma de corriente o del interruptor. - Motor bloqueado - Cable desenchufado del interruptor. - Cable del condensador desconectado - Condensador interrumpido	- Intente encender la unidad con otro cable de alimentación - Reemplace los fusibles; Si hay otra intervención mientras enciende la unidad, desmonte el equipo y conecte todos los cables al motor. El motor está en cortocircuito, entonces reemplácelo. - Abra el dispositivo y vuelva a enchufar los cables. - Reemplazar el motor - Vuelva a conectar el cable al interruptor - Reconecte el cable del capacitor - Reemplace el condensador
NO HAY SUCCION: - Controla el BOWL y la CUBIERTA. - Controlar el motor - Controlar el sistema neumático	- Desconecte todas las tuberías de accesorios del equipo; Encienda el interruptor, cierre la tubería con 0,80 bar. Si la exposición está disponible, el defecto está en los accesorios (en el otro caso vea el punto 5). - Si el problema ocurre con el vacío, es necesario desmontar el carenado superior, desconectar la tubería del motor y verificar la succión del motor con el toque (con un dedo). Tenga cuidado de NO TOCAR los cables y los contactos eléctricos (en caso de succión normal, consulte el punto 6). - Si durante el paso 5. el motor funciona bien, es necesario controlar el circuito neumático (tuberías, unión de tuberías, regulador, tubería gruesa). Verificar su fijación e integridad.	- Asegúrese de que los accesorios no estén rotos o agrietados y verifique el ensamblaje correcto; - Asegúrese de que el filtro antibacteriano no esté obstruido. - Reemplace el motor: un uso erróneo del producto podría haber causado la infiltración de moco en el motor, obstruyéndolo. - Repáre eventuales partes móviles y reemplace las defectuosas. Es posible detectar defectos debidos a incrustaciones de moco .
El aparato es muy ruidoso	Rotura o desconexión de componentes internos.	Restaura la fijación o sustituya las piezas defectuosas y / o rotas.
EL PRODUCTO NO RECARGA (solo para equipos con batería)	- Tarjeta interrumpida - Clip de carga desconectado	- Reemplace la tarjeta (posible cortocircuito dos a una infiltración de moco) - Vuelva a conectar el clip de carga

MAIN SPARE PARTS AVAILABLE

In the table we report all the spare parts supplied by CA-MI for their replacement in case of failure, wear and tear or damage

SPARE PART	DESCRIPTION OF SPARE PART	REF #
PUMP SET	It includes all the already assembled pneumatic parts of the electro-pump and the components which return the mechanical force coming from the electric motor ready to be connected to the electric motor.	SP 0022
MOTOR SET	It includes all the electric parts of the electro-pump, the mobile fixing system and the capacitor, already assembled and ready to be connected to the pneumatic part of the motor and then fixed on the mobile itself.	SP 0023/01
WHOLE ELECTRO-PUMP	The spare part includes the whole already assembled electro-pump, complete of pneumatic parts (whole pump set) and of electric parts (electric motor and components which return the mechanical force), and with the capacitor and the silencer, too.	MA 310100/02
REGULATOR UNIT	It includes not only the whole already assembled regulator, but also the whole fastener, in order to fix it to the device's fairing	SP 0006/03
CAPACITOR	It only includes the capacitor.	SP 0024
SWITCH	It includes only the switch with its cover. The packaging contains n°2 complete kits	SP 0009/04
INTERNAL PIPES	It includes a stretch of pipe skein long enough to replace every part of the internal pneumatic circuit.	SP 0026/04
FUSE CARRIER SOCKET	It includes the power socket (no fuse).	SP 0086/02
VACUUM METRE UNIT	It includes the measurement tool and the whole fastener to fix it to the device's fairing	SP 0017
HANDLE	It only includes the handle	SP 0014/01
DEVICE'S WHOLE BOTTOM	It includes the device's bottom with pre-assembled wheels.	SP 0134
POWER CORD	It only includes the power cord.	SP 0021
ANTI-VIBRATION MOTOR KIT	It only includes the anti-vibration motor component. The packaging contains n°3 pieces	SP 0025/02
COMPLETE 1000cc JAR	It includes the 1000ml jar with the cover with "too full" valve, all sterilizable. The bowò is already assembled and ready to be used.	RE 210001
JAR'S LID	It includes the cover, with "too full" valve, all sterilizable. The cover is already assembled and ready to be used.	RE 210352

SPARE PART	DESCRIPTION OF SPARE PART	REF #
1000ml JAR	It includes the 1000ml whole bowl , sterilizable with autoclave. The jar is ready to be used.	RE 210003
COMPLETE 2000cc JAR	It includes the 2000ml bowl with the cover with “too full” valve, all sterilizable. The jar is already assembled and ready to be used.	RE 210351
2000cc JAR	It includes the 2000ml whole bowl , sterilizable with autoclave.	RE 210353
SUCTION KIT (WITHOUT FILTER)	It includes the transparent silicone pipes kit to connect the bowl to the external connector and the feeler pipe fitting. Antibacterial filter NOT included..	RE 210355/02
SUCTION KIT (WITH FILTER)	It includes the transparent silicone pipes for the connections, bowl-filter-device and n°1 antibacterial filter.	SP 0036/01
ANTIBACTERIAL FILTER	The packaging includes only the filter. (The code refers to the single component).	SP 0046
ANTIBACTERIAL FILTER	The packaging includes only the filter. (The code refers to the pack of 10pcs)	SP 0046/01
SUCTION PROBE	It only includes the probe (10pz) in original sterile package.	RE 210400
CONICAL UNION FOR SUCTION PROBES	It only includes the conical union.	RE 210410

N.B:

Several other accessories are available as well, as indicated in the CA-MI price list.

REPLACING THE PUMP UNIT

All the references indicated below refer to the FIG.7 p.29 exploded view

Required equipment

- Electric/pneumatic screwdriver with drills for cross-shaped incision (or cross-shaped drill screwdriver)
- Pliers
- CH7 Hexagonal spanner
- CH3 Hexagonal spanner

REPAIRING STEPS

Mobile opening

A. Unplug the device from the electrical outlet.

- B. Position the device on a plane, dry surface, (covered with anti-slip and anti-scratch material if possible) so that the control panel will be directed to the user.
- C. Position the device upside-down and unscrew the 4 screws (POS. 57) on the bottom which fix the superior panel (POS. 44) to the bottom (POS. 53). Open the device, make sure not to touch the electric and pneumatic connections which connect the components on the bottom and on the device and don't permit the complete extraction of the mobile itself..

Disconnecting electric and pneumatic connections

- A. Cut the bands which tighten the internal cables with each other, unplug the two cables coming from the motor (POS. 25) from the switch (POS. 46) placed on the control panel.
- B. Take with the hands or with pliers the 6x10 transparent silicon tube (POS. 29) which connects the suction pipe union of the pump unit and disconnect it leaving it connected to the pneumatic circuit.
- C. Disconnect from the other end of the head the pipe with the silencer 8 (POS. 27 – 28 – 29) making sure not to damage it, in order to use it again with the pump. Now the mobile is disconnected from the other parts of the device and it can be placed apart; the electro-pump is then free from electric or pneumatic connections

Extracting the electro-pump and replacing the PUMP UNIT.

- A. Using a CH7 hexagonal spanner, unscrew the 4 screws (POS. 2) which lock the anti-vibrants supports (POS. 1) of the plate (POS. 4) to the bottom itself, working from under the device.
- B. Now the motor is free: pull it apart from the bottom.
- C. Position the motor on the working table: the fan must be at the right side of the user and the head at the opposite side of the user. Now the user can see the chamber of the connecting rod from which the user can operate on the motor eccentric.
- D. With a hand rotate the motor fan until the chamber of the grain (POS. 10) of fixing of the eccentric (POS. 11) is well visible, with CH4 spanner unscrew the grain and extract from the left side the connecting rod/ eccentric system from the motor tree.
- E. Now position the motor with the head at the side of the user and unscrew the 4 fixing screws of the pump unit (POS. 20). Now extract the whole pump unit (from POS. 8 to POS. 20).
- F. Insert the new pump unit operating as follows:
- Insert the complete piston (from POS. 8 to POS. 14), inserting the eccentric on the hole left free after the extraction of the damaged pump unit, now insert the motor tree in the hole of the eccentric and clock it tighten the grain of the eccentric.
 - Insert the cylinder (POS. 15) in the circular hole of the pump support; the side of the cylinder with the blunting must remain on the system. Insert in the cylinder the piston
 - Take the cylinder support (POS. 17) and insert the membrane (POS. 18) blocking the two holes in the two cylinder of reference. Insert O-RING (POS. 16) on the opposite side of the support.
 - Block the side with the O-ring of the cylinder support on the part of the cylinder which is visible from the pump support and insert the head (POS. 19) on the part of the cylinder support which remains external, so that the holes of the head are on the references of the cylinder support.

- When the head is well blocked, rotate the pipe union on the same direction of the electric motor, then block the whole system screwing on the pump support the 4 fixing screws (POS. 20).

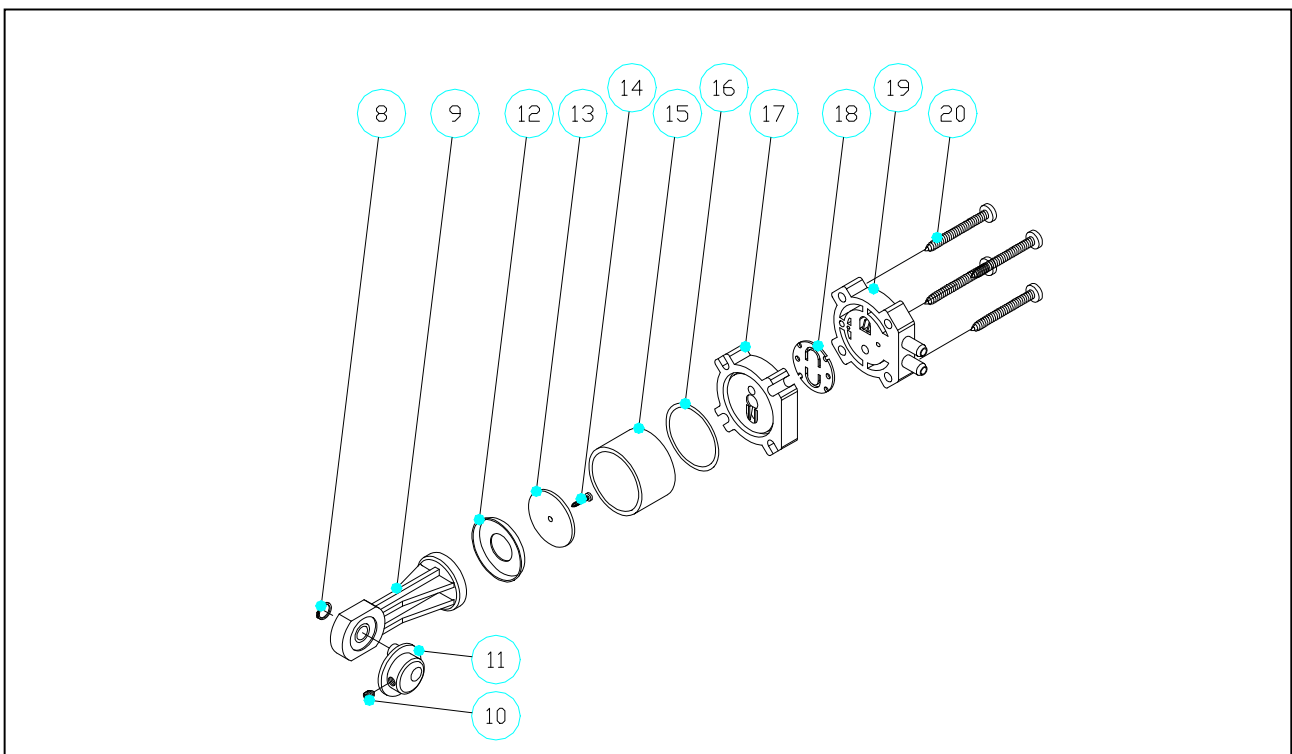
Restoring the device

- Position the electro-pump on the bottom, insert the anti-vibrants in the holes on the bottom itself. The pump orientation is determined by the holes on the bottom. From under the bottom, screw, on the anti-vibrants visible, the three fixing screws (POS. 63) and tighten strongly in order to block the motor.
- Place the motor next to the bottom and restore the electric circuit (electric scheme on page 28) connecting the ends of the black and blue motor cables previously unplugged to the ends left free on the switch (POS. 46). Make sure that the cables are still well tighten together with a band.
- Restore the pneumatic circuit connecting again the air pipe left free in the device to the suction pipe of the head. Insert again the silencer with the pipe previously extracted on the other end of the head and fix it as it was before using a specific band (POS. 61)
- Place again the mobile on the bottom in the final position and before closing definitely the device, control the success of intervention connecting it to the power source, turning it on and verifying that the maximum suction (obtained closing the suction pipe end) is equal or more than -0.80 bar (visible on the vacuum metre).
- If everything's ok, unplug the device from the power source again and close the covering by screwing the 4 fixing screws (POS. 54), giving attention to the fuse carrier socket (POS 54) which has to be inserted in the hole created between the bottom (POS. 53) and the superior plate (POS. 44) of the device. On the contrary, if the device doesn't turn on, if it is too noisy or presents some troubles with the value of maximum pressure, unplug it from the power source and control that the electric and pneumatic connections and the camping of the internal parts are all entire. If the trouble persists control the device, as described in the table on page 7 and/or contact CA-MI technical service.

NOTES

CA-MI manufacturing always recommends to replace the whole internal pipes set which constitutes the pneumatic circuit, whenever such an intervention is required. This replacement is necessary to restore totally the suction circuit efficiency. For this reason specific replacement kits are available.

Fig.1 – Components of the Pump Unit



REPLACING THE MOTOR UNIT

All the references indicated below refer to the FIG.7 p.29 exploded v

Required equipment

- Electric/pneumatic screwdriver with drills for cross-shaped incision (or cross-shaped drill screwdriver)
- Pliers
- CH7 Hexagonal spanner
- CH3 Allen spanner

REPAIRING STEPS

Mobile opening

- Unplug the device from the electrical outlet.**
- Position the device on a plane, dry surface, (covered with anti-slip and anti-scratch material if possible) so that the control panel will be directed to the user.
- Position the device upside-down and unscrew the 4 screws (POS. 57) on the bottom which fix the superior panel (POS. 44) to the bottom (POS. 53). Open the device, make sure not to touch the electric and pneumatic connections which connect the components on the bottom and on the device and do not permit the complete extraction of the mobile itself.

Distacco connessioni elettriche e pneumatiche

- Cut the bands which tighten together the internal cables, then unplug the two thin cables coming from the motor (POS. 25) from the contacts of the switch (POS. 46) placed on the control panel (POS. 52).
- Take the 6x10 transparent silicone pipe (POS.29) which connect to the suction pipe end with the hands or with pliers and disconnect it, leaving it still connected to the remaining pneumatic circuit..
- Disconnect also the pipe with silencer system (POS. 27 - 28 - 29) from the other end of the head; make sure not to damage it during this operation so that it can be used again with the new pump unit. Now the mobile is separated from the device and it be placed apart until it will be mounted again on the bottom of the mobile and the electro-pump is free from electric and pneumatic connections.

Extracting the electro-pump and replacing the MOTOR UNIT.

- Using a CH7 hexagonal spanner, unscrew the 4 screws (POS. 2) which lock the anti-vibrants supports (POS. 1) of the plate (POS. 4) to the bottom itself, working from under the device.
- Now the motor is free: pull it apart from the bottom.
- Position the motor on the working table: the fan must be at the right side of the user and the head at the opposite side of the user. Now the user can see the chamber of the connecting rod from which the user can operate on the motor eccentric.
- With a hand rotate the motor fan until the chamber of the grain (POS. 10) of fixing of the eccentric (POS. 11) is well visible, with CH4 spanner unscrew the GRANO and extract from the left side the connecting rod/eccentric system from the motor tree.
- Now position the motor with the head at the side of the user and unscrew the 4 fixing screws of the pump unit (POS. 20). Now extract the whole pump unit (from POS. 8 to POS. 20).
- Insert the new pump unit operating as follows:
 - Insert the complete piston (from POS. 8 to POS. 14), inserting the eccentric on the hole left free after the extraction of the damaged pump unit, now insert the motor tree in the hole of the eccentric and clock it tighten the grain of the eccentric.
 - Insert the cylinder (POS. 15) in the circular hole of the pump support; the side of the cylinder with the blunting must remain on the system. Insert in the cylinder the piston.
 - Take the cylinder support (POS. 17) and insert the membrane (POS. 18) blocking the two holes in the two cylinder of reference. Insert O-RING (POS. 16) on the opposite side of the support.
 - Block the side with the O-ring of the cylinder support on the part of the cylinder which is visible from the pump support and insert the head (POS. 19) on the part of the cylinder support which remains external, so that the holes of the head are on the references of the cylinder support.
 - When the head is well blocked, rotate the pipe union on the same direction of the electric motor, then block the whole system screwing on the pump support the 4 fixing screws (POS. 20).

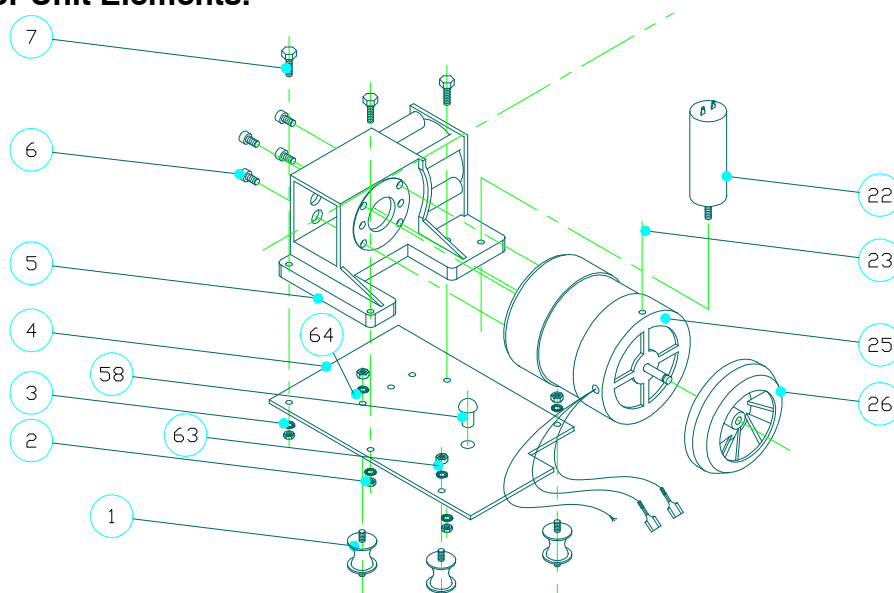
Restoring the device

- A. Position the electro-pump on the bottom, insert the anti-vibrants in the holes on the bottom itself. The pump orientation is determined by the holes on the bottom. Using 3 M6 screws and 3 M6 toothed washers (POS. 62 – 63) tighten the anti-vibrants to the plate. From under the bottom, screw, on the anti-vibrants visible, the three fixing screws (POS. 63) and tighten strongly in order to block the motor.
- B. Place the motor next to the bottom and restore the electric circuit (electric scheme on page 28) connecting the ends of the black and blue motor cables previously unplugged to the ends left free on the switch (POS. 46). Make sure that the cables are still well tighten together with a band.
- C) Restore the pneumatic circuit connecting again the air pipe left free in the device to the suction pipe of the head. Insert again the silencer with the pipe previously extracted on the other end of the head and fix it as it was before using a specific band (POS. 61)
- D) Place again the mobile on the bottom in the final position and before closing definitely the device, control the success of intervention connecting it to the power source, turning it on and verifying that the maximum suction (obtained closing the suction pipe end) is equal or more than -0.80 bar (visible on the vacuum metre).
- E) If everything's ok, unplug the device from the power source again and close the covering by screwing the 4 fixing screws (POS. 54), giving attention to the fuse carrier socket (POS 54) which has to be inserted in the hole created between the bottom (POS. 53) and the superior plate (POS. 44) of the device. On the contrary, if the device doesn't turn on, if it is too noisy or presents some troubles with the value of maximum pressure, unplug it from the power source and control that the electric and pneumatic connections and the camping of the internal parts are all entire. If the trouble persists control the device, as described in the table on page 7 and/or contact CA-MI technical service.

NOTES

- CA-MI manufacturing always recommends to replace the whole internal pipes set which constitutes the pneumatic circuit, whenever such an intervention is required. This replacement is necessary to restore totally the suction circuit efficiency. For this reason specific replacement kits are available.
- Considering the cost of the material and of the technical service, CA-MI recommends the replacing of the whole electro-pump and not only of the motor unit, in case of damages on the electro-mechanical part of the motor.

Fig.2 – Motor Unit Elements.



REPLACING THE WHOLE ELECTROPUMP

All the references indicated below refer to the FIG. 7 p. 29 exploded view

Required equipment

- Electric/pneumatic screwdriver with drills for cross-shaped incision (or cross-shaped drill screwdriver)
- Pliers
- CH10 hexagonal spanner

REPAIRING STEPS

Mobile opening

- Unplug the device from the electrical outlet.**
- Position the device on a plane, dry surface, (covered with anti-slip and anti-scratch material if possible) so that the control panel will be directed to the user.
- Position the device upside-down and unscrew the 4 screws (POS. 57) on the bottom which fix the superior panel (POS. 44) to the bottom (POS. 53). Open the device, make sure not to touch the electric and pneumatic connections which connect the components on the bottom and on the device and do not permit the complete extraction of the mobile itself.

Detachment of electrical and pneumatic connections

- Cut the bands which tighten together the internal cables, then unplug the two thin cables coming from the motor (POS. 25) from the contacts of the switch (POS. 46) placed on the control panel (POS. 52).
- Take the transparent silicone pipe (POS. 29) which connects to the suction pipe end with the hands or with pliers and disconnect it, leaving it still connected to the remaining pneumatic circuit..
- Disconnect also the pipe with silencer system (POS. 27 - 28 - 29) from the other end of the head; make sure not to damage it during this operation so that it can be used again with the new pump unit. Now the mobile is separated from the device and it can be placed apart until it will be mounted again on the bottom of the mobile and the electro-pump is free from electric and pneumatic connections

Extracting and replacing the WHOLE ELECTRO-PUMP

- Using a CH7 hexagonal spanner, unscrew the 3 nuts (POS. 63) which lock the anti-vibrants supports (POS. 1) of the motor to the bottom, working from under the device (POS. 53). Unscrew and extract the fixing elements of the electro-pump.
- Now the motor is free: pull it apart from the bottom
- Take the whole replacing electro-pump and position it on the bottom, inserting the anti-vibrants in the holes on the bottom itself. The pump orientation is determined by the holes on the bottom. From under the bottom, screw, on the anti-vibrants visible, the three fixing screws (POS. 63) and tighten strongly in order to block the motor.

Restoring the device

- Position the electro-pump on the bottom, insert the anti-vibrants in the holes on the bottom itself. The pump orientation is determined by the holes on the bottom. Using 3 M6 screws and 3 M6 toothed washers (POS. 62 – 63) tighten the anti-vibrants to the plate. From under the bottom, screw, on the anti-vibrants visible, the three fixing screws (POS. 63) and tighten strongly in order to block the motor.
- Place the motor next to the bottom and restore the electric circuit (electric scheme on page 28) connecting the ends of the black and blue motor cables previously unplugged to the ends left free on the switch (POS. 46). Make sure that the cables are still well tightened together with a band.
- Restore the pneumatic circuit connecting again the air pipe left free in the device to the suction pipe of the head. Insert again the silencer with the pipe previously extracted on the other end of the head and fix it as it was before using a specific band (POS. 61)
- Place again the mobile on the bottom in the final position and before closing definitely the device, control the success of intervention connecting it to the power source, turning it on and verifying that the

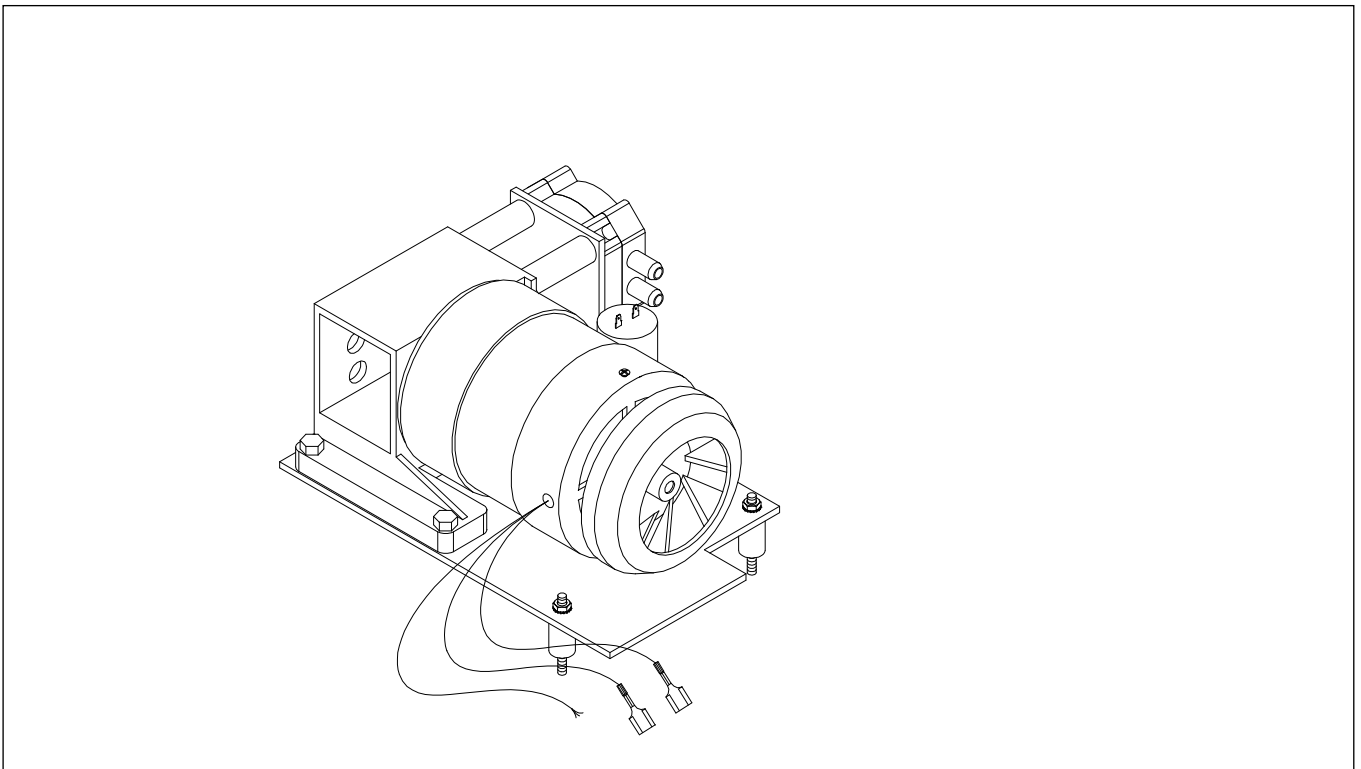
maximum suction (obtained closing the suction pipe end) is equal or more than -0.80 bar (visible on the vacuum metre).

- E. If everything's ok, unplug the device from the power source again and close the covering by screwing the 4 fixing screws (POS. 54), giving attention to the fuse carrier socket (POS 54) which has to be inserted in the hole created between the bottom (POS. 53) and the superior plate (POS. 44) of the device. On the contrary, if the device doesn't turn on, if it is too noisy or presents some troubles with the value of maximum pressure, unplug it from the power source and control that the electric and pneumatic connections and the camping of the internal parts are all entire. If the trouble persists control the device, as described in the table on page 7 and/or contact CA-MI technical service

NOTES

CA-MI manufacturing always recommends to replace the whole internal pipes set which constitutes the pneumatic circuit, whenever such an intervention is required. This replacement is necessary to restore totally the suction circuit efficiency. For this reason specific replacement kits are available

Fig.3 – Whole Electro-pump



REPLACING ELECTROPUMP ANTI-VIBRANT SUPPORTS

All the references indicated below refer to the FIG. 7 p. 29 exploded view

Required equipment

- Electric/pneumatic screwdriver with drills for cross-shaped incision (or cross-shaped drill screwdriver)
- Pliers
- CH7 hexagonal spanners

REPAIRING STEPS

Mobile opening

A. Unplug the device from the electrical outlet.

- B Position the device on a plane, dry surface, (covered with anti-slip and anti-scratch material if possible) so that the control panel will be directed to the user.
- C Position the device upside-down and unscrew the 4 screws (POS. 57) on the bottom which fix the superior panel (POS. 44) to the bottom (POS. 53). Open the device, make sure not to touch the electric and pneumatic connections which connect the components on the bottom and on the device and don't permit the complete extraction of the mobile itself.

Disconnecting electric and pneumatic connections

In order to replace the bottom it is not necessary to operate on the electric and/or pneumatic components. It is only necessary to take care, during the extraction of the pump from the damaged bottom and during the insertion of it on the spare component, not to touch the wiring so that they could be partially or totally unplugged.

Extracting and replacing the ANTI-VIBRANT SUPPORTS

- A. Using a CH7 hexagonal spanner, unscrew the 3 nuts (POS. 63) which lock the anti-vibrant supports (POS. 1) of the motor to the bottom, working from under the device (POS. 53). Unscrew and extract the fixing elements of the electro-pump.
- B. Now the motor is free: pull it apart from the bottom
- C. Rotate the electro-pump (do not unplug the internal wiring cables) in order to verify which ones (and how many) anti-vibrants (POS. 1) are damaged. Place the pump on a straight surface and, using a CH7 spanner, extract the fixing elements nuts and washers (POS. 63 – 64) of the damaged parts. Extract the damaged anti-vibrants and replace them with the spare parts, fixing again them with M6 nuts and M6 toothed washers.
- D. Take the whole replacing electro-pump and position it on the bottom, inserting the anti-vibrants in the holes on the bottom itself. The pump orientation is determined by the holes on the bottom. From under the bottom, screw, on the anti-vibrants visible, the three fixing screws (POS. 63) and tighten strongly in order to block the motor.

Restoring the device

- A. Place again the mobile on the bottom in the final position and before closing definitely the device, control the success of intervention connecting it to the power source, turning it on and verifying that the maximum suction (obtained closing the suction pipe end) is equal or more than -0.80 bar (visible on the vacuum metre).
- B. If everything's ok, unplug the device from the power source again and close the covering by screwing the 4 fixing screws (POS. 54), giving attention to the fuse carrier socket (POS. 54) which has to be inserted in the hole created between the bottom (POS. 53) and the superior plate (POS. 44) of the device. On the contrary, if the device doesn't turn on, if it is too noisy or presents some troubles with the value of maximum pressure, unplug it from the power source and control that the electric and pneumatic connections and the camping of the internal parts are all entire. If the trouble persists control the device, as described in the table on page 7 and/or contact CA-MI technical service

REPLACING THE REGULATOR UNIT

All the references indicated below refer to the FIG. 7 p.29 exploded view

Required equipment

- Electric/pneumatic screwdriver with drills for cross-shaped incision (or cross-shaped drill screwdriver)
- Pliers
- CH17 hexagonal spanner

REPAIRING STEPS

Mobile opening

A. Unplug the device from the electrical outlet.

- B Position the device on a plane, dry surface, (covered with anti-slip and anti-scratch material if possible) so that the control panel will be directed to the user.
- C. Position the device upside-down and unscrew the 4 screws (POS. 57) on the bottom which fix the superior panel (POS. 44) to the bottom (POS. 53). Open the device, make sure not to touch the electric and pneumatic connections which connect the components on the bottom and on the device and don't permit the complete extraction of the mobile itself.

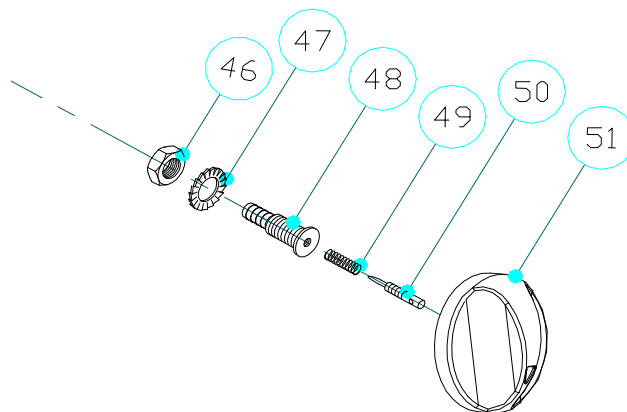
Disconnecting the pneumatic connections and replacing the REGULATOR UNIT

- A. Take the transparent silicone pipe (POS.29) which connect to the regulator connector (POS. 48) which is visible internally from the control panel (POS. 52) with the hands or with pliers and disconnect it, leaving it still connected to the remaining pneumatic circuit..
- B. With the hexagonal CH17 spanner release the nut with the washer (POS. 47 and 66) which tighten the regulator to the mobile. Unscrew and extract the fixing elements and the regulator.
- C. Take the new whole regulator unit (from POS. 48 to 51) and insert it in the hole left free on the control panel (POS. 52); the arrow on the handle must be placed downside, on the right side (about 4 hours); block the regulator to the firing fixing from the inside with a nut and a screw.

Restoring the device

- A. Restore the pneumatic circuit connecting again the air pipe left free in the device to the suction pipe of the head. Insert again the silencer with the pipe previously extracted on the other end of the head and fix it as it was before using a specific band (POS. 61)
- B. Place again the mobile on the bottom in the final position and before closing definitely the device, control the success of intervention connecting it to the power source, turning it on and verifying that the maximum suction (obtained closing the suction pipe end) is equal or more than -0.80 bar (visible on the vacuum metre).
- C. If everything's ok, unplug the device from the power source again and close the covering by screwing the 4 fixing screws (POS. 54), giving attention to the fuse carrier socket (POS 54) which has to be inserted in the hole created between the bottom (POS. 53) and the superior plate (POS. 44) of the device. On the contrary, if the device doesn't turn on, if it is too noisy or presents some troubles with the value of maximum pressure, unplug it from the power source and control that the electric and pneumatic connections and the camping of the internal parts are all entire. If the trouble persists control the device, as described in the table on page 7 and/or contact CA-MI technical service

Fig.4 – Regulator Unit Elements



REPLACING THE CAPACITOR

All the references indicated below refer to the FIG. 7 p. 29 exploded view

Required equipment

- Electric/pneumatic screwdriver with drills for cross-shaped incision (or cross-shaped drill screwdriver)
- Pliers
- Electric solder
- Tin

REPAIRING STEPS

Mobile opening

A. Unplug the device from the electrical outlet.

- B Position the device on a plane, dry surface, (covered with anti-slip and anti-scratch material if possible) so that the control panel will be directed to the user.
- C Position the device upside-down and unscrew the 4 screws (POS. 57) on the bottom which fix the superior panel (POS. 44) to the bottom (POS. 53). Open the device, make sure not to touch the electric and pneumatic connections which connect the components on the bottom and on the device and do not permit the complete extraction of the mobile itself..

Unplugging the electric connections and replacing the CAPACITOR

- A. Cut the bands which tighten the thin cables together and unplug the motor cable ends (POS.25) from the switch (POS. 46).
- B. Unplug from the damaged capacitor (POS. 22) the ends of the blue and red thin cables, taking care of not touching other internal parts.
- C. With the hands or with pliers rotate anti-clockwise the capacitor and unscrew it until it will be free from the pump sport (POS. 5)
- D. Insert the spare capacitor in the hole of the motor support from which it has been extracted the damaged element and screw it until it will be well fixed.

Restoring the device

- A. Restore the electric circuit (electric scheme on pag. 28) connecting the ends of the blue and red cables coming from the motor (POS. 25) to the spare part and connect the ends of the motor black and blue cables to the ends left free on the switch. Make sure that the cables are all strongly tightened together with the specific bands.
- B. Place again the mobile on the bottom in the final position and before closing definitely the device, control the success of intervention connecting it to the power source, turning it on and verifying that the maximum suction (obtained closing the suction pipe end) is equal or more than -0.80 bar (visible on the vacuum metre).
- C. If everything's ok, unplug the device from the power source again and close the covering by screwing the 4 fixing screws (POS. 54), giving attention to the fuse carrier socket (POS 54) which has to be inserted in the hole created between the bottom (POS. 53) and the superior plate (POS. 44) of the device. On the contrary, if the device doesn't turn on, if it is too noisy or presents some troubles with the value of maximum pressure, unplug it from the power source and control that the electric and pneumatic connections and the camping of the internal parts are all entire. If the trouble persists control the device, as described in the table on page 7 and/or contact CA-MI technical service



During this operation make sure not short circuit the capacitor contacts with the hands or with any other tool.

REPLACING THE SWITCH

All the references indicated below refer to the FIG. 7 p.30 exploded view

Required equipment

- Electric/pneumatic screwdriver with drills for cross-shaped incision (or cross-shaped drill screwdriver)
- Small flat-head screwdriver for intervention on electric elements

REPAIRING STEPS

Mobile opening

- Unplug the device from the electrical outlet.**
- Position the device on a plane, dry surface, (covered with anti-slip and anti-scratch material if possible) so that the control panel will be directed to the user.
- Position the device upside-down and unscrew the 4 screws (POS. 57) on the bottom which fix the superior panel (POS. 44) to the bottom (POS. 53). Open the device, make sure not to touch the electric and pneumatic connections which connect the components on the bottom and on the device and don't permit the complete extraction of the mobile itself.

Unplugging electric connections and replacing the SWITCH

- Unplug the ends of the internal wiring cables from the damaged component, leaving the cables still tighten together with the bands.
- From the inside of the motor, use a flat-headed screwdriver on the anchoring levers of the switch (POS. 46) so that it could be extracted from the control panel (POS. 52).
- Insert the spare switch on the hole left free on the control panel, so that the I symbol will be directed to the vacuum metre unit (POS. 45). Make sure that the anchoring levers will be well fixed.

Restoring the device

- Restore the electric circuit (electric scheme on pag. 28) connecting again the internal electric wiring cables to the switch as follows:
 - Connect the two thin cables coming from the power plug (POS. 55) to the switch contacts in the **O** position
 - Connect the thin cables coming from the motor (POS. 25) and from the capacitor to the switch contacts in the **I** position.Always make sure that the cables are well tighten together with the specific bands.
- Place again the mobile on the bottom in the final position and before closing definitely the device, control the success of intervention connecting it to the power source, turning it on and verifying that the maximum suction (obtained closing the suction pipe end) is equal or more than -0.80 bar (visible on the vacuum metre).
- If everything's ok, unplug the device from the power source again and close the covering by screwing the 4 fixing screws (POS. 54), giving attention to the fuse carrier socket (POS 54) which has to be inserted in the hole created between the bottom (POS. 53) and the superior plate (POS. 44) of the device. On the contrary, if the device doesn't turn on, if it is too noisy or presents some troubles with the value of maximum pressure, unplug it from the power source and control that the electric and pneumatic connections and the camping of the internal parts are all entire. If the trouble persists control the device, as described in the table on page 7 and/or contact CA-MI technical service

REPLACING THE FUSE CARRIER SOCKET

All the references indicated below refer to the FIG. 7 p. 29 exploded view

Required equipment

- Electric/pneumatic screwdriver with drills for cross-shaped incision (or cross-shaped drill screwdriver)
- Pliers
- Electric solder
- Tin

REPAIRING STEPS

Mobile opening

- Unplug the device from the electrical outlet.**
- Position the device on a plane, dry surface, (covered with anti-slip and anti-scratch material if possible) so that the control panel will be directed to the user.
- Position the device upside-down and unscrew the 4 screws (POS. 57) on the bottom which fix the superior panel (POS. 44) to the bottom (POS. 53). Open the device, make sure not to touch the electric and pneumatic connections which connect the components on the bottom and on the device and do not permit the complete extraction of the mobile itself.

Unplugging the electric connections and replacing the FUSE CARRIER SOCKET

- Unplug the ends of the two thin cables coming from the switch (POS. 46), placed on the control panel (POS. 52), taking them still tighten together with the specific band.
- Now it is possible to replace the damaged component with the spare part.

Restoring the device

- Restore the electric circuit (electric scheme on pag. 28) connecting again the internal electric wiring cables of the switch as follows:
- The two switch thin cables (POS.46), placed on the control panel (POS. 52) must be connected to the two contacts of the fuse carrier socket.
- Always make sure that the cables are well tighten together with the specific bands.
- Open the fuse carrier socket chamber and insert 1 glass fuse (5x20) F1.6A
- Place again the mobile on the bottom in the final position and before closing definitely the device, control the success of intervention connecting it to the power source, turning it on and verifying that the maximum suction (obtained closing the suction pipe end) is equal or more than -0.80 bar (visible on the vacuum metre).
- If everything's ok, unplug the device from the power source again and close the covering by screwing the 4 fixing screws (POS. 54), giving attention to the fuse carrier socket (POS 54) which has to be inserted in the hole created between the bottom (POS. 53) and the superior plate (POS. 44) of the device. On the contrary, if the device doesn't turn on, if it is too noisy or presents some troubles with the value of maximum pressure, unplug it from the power source and control that the electric and pneumatic connections and the camping of the internal parts are all entire. If the trouble persists control the device, as described in the table on page 7 and/or contact CA-MI technical service

REPLACING THE VACUUM METRE UNIT

All the references indicated below refer to the FIG. 7 p. 29 exploded view

Required equipment

- Electric/pneumatic screwdriver with drills for cross-shaped incision (or cross-shaped drill screwdriver)
- Pliers
- CH18 hexagonal spanner

REPAIRING STEPS

Mobile opening

A. Unplug the device from the electrical outlet.

B Position the device on a plane, dry surface, (covered with anti-slip and anti-scratch material if possible) so that the control panel will be directed to the user.

C. Position the device upside-down and unscrew the 4 screws (POS. 57) on the bottom which fix the superior panel (POS. 44) to the bottom (POS. 53). Open the device, make sure not to touch the electric and pneumatic connections which connect the components on the bottom and on the device and do not permit the complete extraction of the mobile itself..

Disconnecting pneumatic connections and replacing the VACUUM METRE UNIT

- Working with the hands or with pliers through the mobile's hole, left free after the air grating extraction, take up the 6x10 transparent silicone pipes (POS.29) linked up with the vacuum metre connector (POS. 45), which appears from the internal of the control panel (POS.447). Disconnect it from the damaged part and leave it connected to the remaining pneumatic circuit.
- Insert the plain-head screwdriver in the hole and use it to unscrew the two vacuum metre connecting rods. Rotate the vacuum metre slightly until it comes off the blocking system and extract it from the mobile part, operating from the outside. Working from the inside hold the blocking system up and extract it.
- Take the new whole vacuum metre unit (POS.45) and insert it in the hole left free on the control panel.

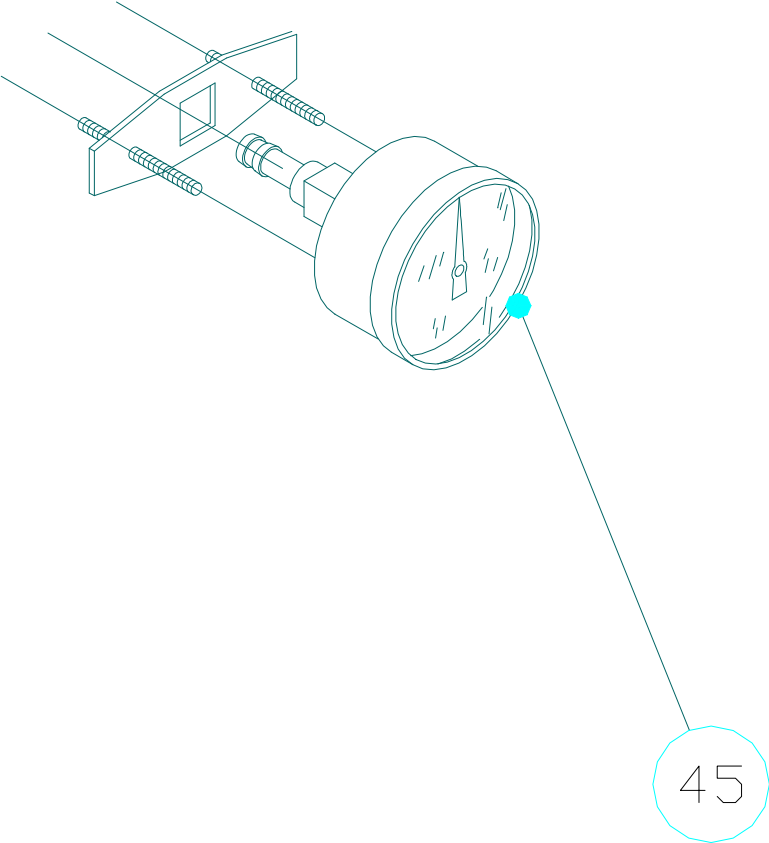
Restoring the device

- Restore the pneumatic circuit reconnecting strongly the air pipe left free to the vacuum metre connector
- Place again the mobile on the bottom in the final position and before closing definitely the device, control the success of intervention connecting it to the power source, turning it on and verifying that the maximum suction (obtained closing the suction pipe end) is equal or more than -0.80 bar (visible on the vacuum metre).
- If everything's ok, unplug the device from the power source again and close the covering by screwing the 4 fixing screws (POS. 54), giving attention to the fuse carrier socket (POS 54) which has to be inserted in the hole created between the bottom (POS. 53) and the superior plate (POS. 44) of the device. On the contrary, if the device doesn't turn on, if it is too noisy or presents some troubles with the value of maximum pressure, unplug it from the power source and control that the electric and pneumatic connections and the camping of the internal parts are all entire. If the trouble persists control the device, as described in the table on page 7 and/or contact CA-MI technical service.

NOTES

CA-MI manufacturing always recommends to replace the whole internal pipes set which constitutes the pneumatic circuit, whenever such an intervention is required. This replacement is necessary to restore totally the suction circuit efficiency. For this reason specific replacement kits are available

Fig.5 – Vacuum Metre Unit Elements



REPLACING THE HANDLE

All the references indicated below refer to the FIG. 7 p.29 exploded view

Required equipment

- A. Electric/pneumatic screwdriver with points/drills for cross-shaped incision (or cross-shaped point screwdriver)
- B. CH10 hexagonal spanner

REPAIRING STEPS

Mobile opening

- A. **Unplug the device from the electrical outlet.**
- B. Position the device on a plane, dry surface, (covered with anti-slip and anti-scratch material if possible) so that the control panel will be directed to the user.
- C. Position the device upside-down and unscrew the 4 screws (POS. 57) on the bottom which fix the superior panel (POS. 44) to the bottom (POS. 53). Open the device, make sure not to touch the electric and pneumatic connections which connect the components on the bottom and on the device and do not permit the complete extraction of the mobile itself.

Replacing the HANDLE

- A. Using the CH10 hexagonal spanner release from the inside the 4 screws (POS. 56) which tighten the handle (POS. 43) to the superior panel (POS. 44) of the device. Unscrew and extract the fixing elements releasing the component to be replaced.
- B. Position the replacing handle and position it so that the holes left free on the superior panel face it; fix the handle on the panel, fixing it from the inside with fixing screws (POS. 56).

Restoring the device

- A. Before closing the mobile verify the blocking of the handle just pulling it.
- B. If everything's OK go on closing the mobile and fixing the firing.

REPLACING THE BOTTOM

All the references indicated below refer to the FIG. 7 p. 29 exploded view

Required equipment

- Electric/pneumatic screwdriver with drills for cross-shaped incision (or cross-shaped drill screwdriver)
- Hexagonal spanner

REPAIRING STEPS

Mobile opening

- Unplug the device from the electrical outlet.**
- Position the device on a plane, dry surface, (covered with anti-slip and anti-scratch material if possible) so that the control panel will be directed to the user.
- Position the device upside-down and unscrew the 4 screws (POS. 57) on the bottom which fix the superior panel (POS. 44) to the bottom (POS. 53). Open the device, make sure not to touch the electric and pneumatic connections which connect the components on the bottom and on the device and do not permit the complete extraction of the mobile itself.

Disconnecting electric and pneumatic connections

In order to replace the bottom it is not necessary operate on the electric and/or pneumatic components. It is only necessary to take care, during the extraction of the pump from the damaged bottom and during the insertion of it on the spare component, not to touch the wiring so that they could be partially or totally unplugged.

Unplugging the electro-pump and replacing the DEVICE'S BOTTOM

- Using the hexagonal spanner release, working under the device (POS. 53), the 3 fixing nuts (POS. 63) which fix the anti-vibrants bolts (POS.1) of the motor to the bottom.
- Now the motor can be separated from the bottom of the device (take care not to damage the internal wiring cables). Insert in the spare part the electro-pump inserting its anti-vibrants in the holes on the new bottom. The pump orientation is determined by the position of the holes on the bottom itself. Working under the bottom screw the 3 fixing nuts (POS. 63) on the anti-vibrants and tighten them strongly in order to fix the motor.

Restoring the device

- Place again the mobile on the bottom in the final position and before closing definitely the device, control the success of intervention connecting it to the power source, turning it on and verifying that the maximum suction (obtained closing the suction pipe end) is equal or more than -0.80 bar (visible on the vacuum metre).
- If everything's ok, unplug the device from the power source again and close the covering by screwing the 4 fixing screws (POS. 54), giving attention to the fuse carrier socket (POS 54) which has to be inserted in the hole created between the bottom (POS. 53) and the superior plate (POS. 44) of the device. On the contrary, if the device doesn't turn on, if it is too noisy or presents some troubles with the value of maximum pressure, unplug it from the power source and control that the electric and pneumatic connections and the camping of the internal parts are all entire. If the trouble persists control the device, as described in the table on page 7 and/or contact CA-MI technical service.

Fig. 6 – NEW ASKIR 30 REF RE 310100/02 ELECTRIC DIAGRAM

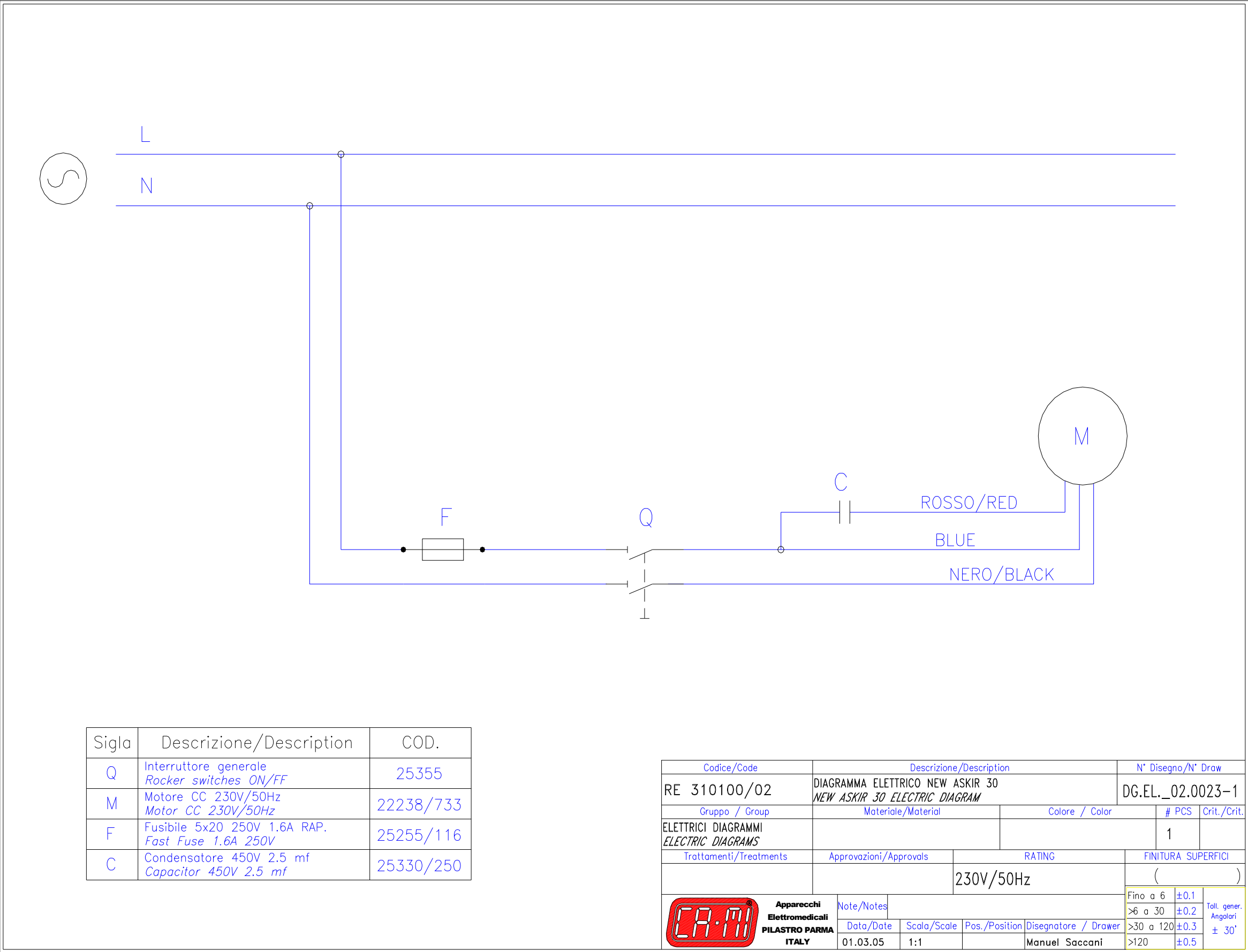


Fig. 7 – NEW ASKIR 30 REF RE 310100/02 (rif. ED.AS._04.0010-2) EXPLODED VIEW

